Design and Evaluation of a Li-Fi Communication System Using LED and Photodiode

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***Abstract*—Visible Light Communication (VLC) is an emerging field in Optical Wireless Communication (OWC) which utilizes the superior modulation bandwidth of Light Emitting Diodes (LEDs) to transmit data.This project explores how a simple Li-Fi link can be built using only LEDs, photodiodes, and Arduino boards. We compared three basic modulation schemes - OOK, FSK, and PWM - to see how well they work under real optical conditions. A framing method with checksums and optical acknowledgments allowed the system to send and confirm messages in both directions without any extra analog circuitry. Our experiments showed that OOK was the most reliable at very short distances, while PWM and FSK struggled more with timing drift and weaker light levels. Overall, the results show what is possible with minimal hardware and point to clear opportunities for improvement using better optics and more accurate timing support.**

I. Introduction

A. Background

Through advancements in LED lighting technology,Visible Light Communication (VLC) has grown rapidly. LEDs from simple illumination sources have become viable carriers of digital information. In contrast to traditional Radio Frequency systems VLC operates using optical wavelengths that enable data transmission through fast intensity changes that remain imperceptible to the human eye. VLC's dual capability of providing illumination and communication makes it well suited for environments where RF usage is restricted or undesirable such as hospitals airplanes and industrial facilities. Because the visible light is not interfered with sensitive equipment and can support a wide bandwidth VLC offers a practical and energy efficient option for short range wireless communication. [1].

B. Motivation

In addition to increasing RF spectrum congestion demand for short-range secure communication and the rapid adoption of IoT systems motivate the exploration of low cost VLC designs. A VLC based on Arduino offers a practical platform for understanding modulation, detection thresholds, channel impairments and physical layer constraints. Using LEDs and photodiodes our motivation is to develop a simple but reliable VLC link capable of transmitting binary data sequences and file messages. As VLC performance is highly dependent on distance, LED intensity and ambient light our project aims to study how these factors influence reliability and modulation performance.

C. Problem Statement

This project began with a simple question: how much of a communication system can be built using nothing more than visible light and very simple electronic parts?Our objective is to construct a functional Li–Fi link using low-cost Arduinos, a pair of LEDs, and photodiodes, while maintaining message transfer and acknowledgements. We also address several issues such as stable bit decoding, frame synchronization, and noise sensitivity during transmission, especially when comparing the modulations (OOK, FSK, and PWM). Through a fully bidirectional design,the project demonstrates how thoughtful protocol design can overcome the physical limitations of minimal hardware.

D. Challenges

Developing a reliable bidirectional VLC system using two LEDs two photodiodes and only basic timing capabilities of the Arduino introduced several practical challenges. Because the forward data channel, Node A to Node B, and the reverse acknowledgment channel, Node B to Node A, relied entirely on variations of light intensity measured through unconditioned analog readings the link was highly sensitive to distance Following are the main difficulties encountered during the project:

- **Limited communication distance:**The limited communication distance proved to be the most restrictive physical restriction. Experiments conducted at separations of 0.1cm, 0.2 cm, 0.3cm and 0.4cm showed that stable decoding occurs only at very close ranges. The LED signal was strong and cleanly distinguishable from noise at 0.1cm but by 0.3cm threshold instability was significantly increased.
- **Low optical output from standard LEDs:** The common 5 mm indicator LEDs used as transmitters provide relatively low intensity. Without focusing optics or amplification, the LED light spreads out quickly, resulting in a rapid loss of intensity as the distance increases. Even small differences between LED and photodiode resulted

in large drops in signal strength and required near contact placement for consistent operation.

- **Photodiode alignment sensitivity:** The photodiodes are highly directional, so accurate decoding required very precise alignment with the transmitting LED. Even minor changes in angle or position of the photodiode caused large variations in the detected signal strength. Also, reflections or strong ambient light sometimes masked the LED output, making calibration more difficult and increasing the chance of errors in the decoding.
- **Ambient light interference:** Indoor lighting, sunlight and reflective surfaces introduced unpredictable fluctuations into the analog readings. Since receivers used fixed thresholds for bit detection changes in ambient light could produce false transitions or mask real ones. Even though a cardboard box helped reduce interference it did not completely eliminate environmental noise.

These challenges directly impacted the achievable communication distance, bit error rate, and overall system robustness.

E. Methods and Algorithms

Our system implements modulations using basic Arduino I/O, LEDs, and bare photodiodes. The goal here is to compare how different modulation schemes behave on similar hardware while supporting reliable message delivery through framing,timing recovery and message acknowledgements [4]

1) Modulation Techniques: Three LED intensity based modulation methods were developed:

a) On Off Keying (OOK): OOK is the simplest form of modulation where data is encoded directly in the presence or absence of light. By turning on the LED a binary '1' is transmitted while a '0' is transmitted by turning it off. The data is sent within a fixed time interval in OOK. [7]

b) Frequency Shift Keying (FSK): FSK encodes information by changing how quickly the LED toggles between on and off. Two different colors of blinks represent binary 0 and 1. The encoder decodes the symbol by counting the number of transitions observed within each bit window.

c) Pulse Width Modulation (PWM): PWM transfers bits by varying the duration of the LEDs on time within a fixed period. A short pulse encodes '0' while a longer pulse encodes '1'. PWM transfers bits by varying the duration of the LED's on time within a fixed bit period.

2) Receiver Algorithms: All decoding is implemented using raw `analogRead()` values :

- **Thresholding:** It distinguishes light vs dark for OOK and PWM.
- **Pulse duration measurement:** It is used for bit classification of PWM.
- **Transition counting:** It is used to decode FSK blinking frequencies.

3) Framing and Link-Layer Protocol: A simple protocol which assures reliable message transfer:

- **Preamble:** A fixed sync byte(0xA5 or 0xAA) marks the start of each frame.
- **Length field:** It defines the number of payload bytes to follow.
- **Checksum:** A 8 bit sum (mod 256) to validate payload integrity. [6].

F. Summary of Implementation and Results

The system was implemented using two Arduino Uno boards, each equipped with a 5 mm LED for transmission and a photodiode with a 10 kΩ pull-up resistor for reception, forming a bidirectional VLC link with optical acknowledgements. Three modulation schemes - OOK, FSK, and PWM - were developed using software-based timing and threshold-driven decoding on raw photodiode readings. A lightweight link-layer protocol incorporating preambles, length fields, checksums, and ACK responses enabled the reliable transfer of short framed messages as well as longer text streams using packetization.

II. System overview and data collection

A. Main Devices and Hardware Used in the System.

Using two Arduino Uno boards each configured as an independent optical transceiver node the bidirectional LiFi communication system was implemented. [5] Every node had a transmitter element composed of a 5,mm visible light LED driven through a 220 Ω current-limiting resistor enabling fast switching required for OOK, PWM and FSK modulation schemes. For the optical reception, each node used a silicon photodiode connected in a reverse-biased resistive configuration with a 10 kΩ pull-up resistor allowing the Arduino's integrated 10-bit analog to digital converter to measure light The photo-diode resistor network produced large and stable voltage differentials between the illuminated and the dark states which was critical for reliable bit classification.

Consequently each Arduino contained both a transmitter (LED) and a receiver (photodiode), enabling full two-way communication in which data frames from NodeA could be recognized optically by Node B. A dedicated alignment channel was constructed using a cardboard enclosure because consumer LEDs exhibit limited luminous intensity and wide beam dispersion. The goal was to reduce ambient noise, constrain the optical path and maintain a consistent separation of approximately 0.2 cm On solderless breadboards the entire communication circuitry was assembled using standard jumper wiring. Collectively these components provided a minimal yet fully functional hardware platform capable of transmitting framed messages, performing checksum verification and exchanging ACK signals through visible light.

B. Main Software Components Developed

The bidirectional LiFi system was developed using embedded C++ in the Arduino IDE along with the built-in Arduino libraries for digital I/O, analog sampling and timing control. Custom firmware for both nodes was written for OOK,PWM and FSK modulation each requiring its own bit-encoding logic timing routines and photodiode decoding with oversampling

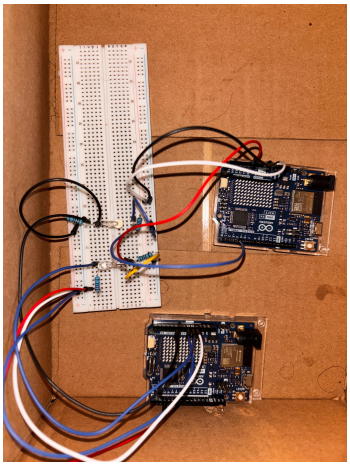

Fig. 1. Actual prototype of the bidirectional Li-Fi system showing both Arduino nodes, LEDs, photodiode receivers, and breadboard wiring inside the cardboard enclosure.

and majority-vote A lightweight framing protocol was designed on top of the physical layer that handles preamble detection header and length parsing checksum verification and message reconstruction . Also an optical ACK / ACK timeout mechanism was implemented so that node B could confirm or reject each received frame by flashing its LED back to node A. It was written manually without external libraries giving us full control over signal timing and allowing software to be tuned to real-world constraints such as LED brightness, photodiode sensitivity and alignment conditions.

C. System Connectivity and Communication Flow.

The system is consists of two Arduino-based LiFi nodes that communicate completely through optical links. Node A sends data frames to Node B while Node B returns the acknowledgement. Each node includes a transmission LED and a photodiode receiver. This is achieved through OOK, PWM or FSK modulation with Node A encoding bits as LED intensity or timing variations and Node B performing continuous photodiode sampling demodulation and checksum validation. Once the frame has been verified, Node B transmits a brief ACK sequence using its LED which Node A detects.This set up provides a complete bidirectional link with independent send and receive paths for each microcontroller.

a) Data Collection Methodology.: Data were collected through repeated experiments performed at controlled LED photodiode distances ranging from 0.1 cm to 0.4 cm. Using each modulation scheme both short test and longer frame messages were transmitted multiple times to observability and bit-level reliability. In order to measure light dark separation, evaluate noise sensitivity and determine appropriate detection thresholds photodiode ADC values were recorded. A total of 20-30 transmissions were executed across all trials to capture variability in alignment ambient light and timing accuracy.

b) Post-Processing and Analysis.: Collected reading and decoding results were analyzed to refine threshold values, adjust bit timing and compare modulation robustness. In order to quantify noise fluctuations oversampled photodiode data were inspected while frame outcomes were used to identify common error patterns especially in longer PWM and FSK transmissions. This analysis led to improvements in the sampling windows, inter-bit gaps and checksum verification leading to a more stable and reliable optical communication protocol. Comparison of OOK, PWM and FSK provided insight into the trade-offs between simplicity noise tolerance and timing sensitivity in low-cost LiFi devices.

III. Methods, Techniques, and Algorithms

In this implemented Visible Light communication (VLC) system, we did the combination of physical layer modulations along with link-layer algorithms to get reliable optical communication using low-cost components like LEDs,photodiodes and Arduino boards.As the proposed system is operating without making any use of the analog amplifier, filters, or hardware synchronization, so the entire system communication stack will revolve around modulation,decoding,framing,error detection, and acknowledgments.All the communication stack is implemented in our system and the below section describes it in detailed way as how all this helped us in building the robust, low cost VLC system which supports the reliable message transfer with different modulation techniques.

A. Modulation Techniques

In this project we implemented and compared all the three modulations (OOK ,FSK and PWM) so that we can understand how each of the them performs on the same hardware under the same channel conditions. [4].

1) On-Off Keying (OOK) modulation: In On-Off Keying (OOK) modulation we are encoding the bits using the simple LED states which are defined as on state of LED is read as 1 and the LED off state is read as 0 which operates on the fixed bit period which is again controlled by the Arduino timing function.On the other hand, the photodiode will detect the readings and compare them against a particular voltage threshold to detect transfer of the bit.Here we implemented the two OOK based method where with each method,we kept on increasing the complexity.

a) Simple OOK Packet and Acknowledgment: In this method we tested all the communication setup by transmitting the basic packet along with acknowledgment to verify the optical path work perfectly.This simple setup allowed us to to validate that the hardware wiring , threshold, LED alignment and the basic and the basic handshake before we move ahead of complex protocols.

- **Sender (Node A):** Here in node A the LED is set on the fixed duration(mostly at 300 ms) which is representing the packet and once a packet is transmitted for the limited window like 2 seconds by sampling it continuously we send its acknowledgment via A0 of Arduino. And if the the photodiode reading is getting droped below the defined threshould then the acknowledgment will considered as received.

- **Receiver (Node B):** now here in this node the photodiode reads the data continuously and when it senses that the value is below the set data threshold it considers that as the packet received and gives the response by flashing the Acknowledgment LED by flashing it for the three times

b) OOK Framed Short Message: In this we used the combination of the header ,length ,checksum and acknowledgment for transmitting the short message by keeping the same setup of connections.

- **Frame Format:** The frame format of the proposed system looks as [HEADER][LENGTH][PAYLOAD BYTES][CHECKSUM] where
  - **HEADER=** 0 x 45 which is chosen for the alternating the pattern which makes it easy to detect the streaming of the bit.
  - **LENGTH=** Here the length is the number of bytes in the payload
  - **CHECKSUM=** The checksum is nothing but the sum of the payload bytes mod 256.
- **Sending (Node A)** Now,each byte is sent in MSB- first from by using the fixed Bit period like 150 ms then the function is defined (like sendbit()) which will simply turn the LED into fully on or off for a particular duration. After the complete frame is send by node A.the node will start watching its own photodiode will wait for for the acknowledgment of the node B LED where it will simply look for the light level to drop below the threshold.Here it will simply take multiple samples just to make sure that their is no false detection and if the good amount of the samples showed that light detection then node A will say that acknowledgment received.
- **Receiving (Node B)** As node B the job is completely opposite like it will simply decode the signal and rebuild the message.Node B will read the bit at a time and shift it into a small 8 bit memory called register so to do it node B simply keep on memory of the register and if it matches 0x45 (1010 0101) then it is considered as a new message is detected and then the next thing will be to see how many characters are their in that message by checking the bytes and read the bytes accordingly. Finally it compares it with checksum send by node A by calculating it as sum of all bytes mod 256 and if it matches,then message is correct and if it does not matches, then message is corrupted. If everything is correct then it will simply send the acknowledgment and Node A sees it using its photo diode and considers the transmission is successful.

c) OOK Long Message Technique: Here in this method our goal is to transmit the long text messages using using OOK modulation so here idea is simple intead of sending the whole message packet we will send the character by character continuously which is inspired from UART technique. so

- **symbol format per character:** here the symbol for format for per character is defined as 1 bit is considered as LED on, each 8 data bit means least significant bit is first and LED off means its a stop bit.
- **Sender (Node A):** So here in this node we reduced the complexity so that we can send each character.and here LED turns on for one bit period and it will tell the node B that its a new string is started now each bit here of the ASCII value is sent LSB first using the regular OOK like 1 for on and 0 for off by making it last for 40 ms.and it will stop the sending when the LED is turned off for one bit period and after this we add a small gap which ultimately helps the receiver to get reset and give better results by avoiding mistakes on the next start bit.This pattern will continiously repeated for each and every charecter in long message which ls inspired form UART technique.
- **Receiver (Node B):** here in node B it does not know when the frame begins so it simply wait for the start bit.now here it wait for LED to get turn on and indicates the start bit then wait for the half bit period which will place node B in middle of the ext bit windows now start reading each 8 data bit so here in every 40 ms sample the photo diode LED on as 1 and LED off as 0 and finally convert the byte into character and immediately print it and finally look for the terminator character so when node B receivers the terminator character it get to know messages finished and it send the acknowledgment back to node A this technique allowed us to send the continuous message to be received without needing the header or lengths.

2) Frequency-Shift Keying (FSK) modulation: Here FSK is the frequency shift keying modulation technique in which we are encoading the digital data by doing the variation in the frequency of the light emmitted by LED.but here in our project FSK is nothing but frequency like behavior using LED blink based on the patterns and pulse timing where we implemented it in multiple ways as

a) Simple FSK Packet and Acknowledgment: Here in this method again we tested the acknowlegement for the current setup of connection using the FSK technique where we used burst based packet detector

- **Sender:** Here in node A we rapidly toggled the LED at a particular fixed short period nearly of 30 ms where the packet duration was 600 ms which ultimately created the distinctive high transition rate optical pattern.
- **Receiver:** so here in node B we are checking the photo diode over a short window of 400 ms and count that how many times the light switched On and off in such a way that if the LED switched states many times then the node B will say that this is the FSK packet and as response it will give back the acknowledgment.

b) FSK Short Message: Here for using the short message we used the pulse -length based fsk steyle scheme to send the actual message as follow

- **Bit encoding** here we used the bit encoding which means how long the LED stays on.in this technique a bit 0 represents short on pulse which lasts for 80ms and then it gets off and for bit 1 represents long on pulse which stays

for 160 ms and then it gets off.so instead of blinking it for fast the length of the time on represent the bit.

- **Frame structure** so here the frame we have set the frame start as 0xAA which is (1010 1010) which has fixed message length of 9 bytes.
- **sender** so here it will start sending the preamble as bits using short pulses and then it sends ASCII bytes of message and then uses the MSB first encoding after completing this it wait for the acknowledgement from the receiver
- **Receiver** Now here the receiver node will wait for until the A0 of aurdino will show the light via led and then it will measure how long the LED's data threshold stays below i.e nothing but on duration and compares it with maximum short or minimum log by deciding 0 or 1 and then it shifts the bits into bytes and make use of sliding window to detect the preamble 0xAA preamble.once it detect the preemble and starts reading the 9 preemble bytes then it prints the message and send the acknowledgment

c) FSK long text message: here in this method we extended the previous short message transmission method by extending pulse length just by changing the frame format very nominal change in the logic. which is explaned as follow

- **Frame format:** here the changed frame format is [PREAMBLE = 0xAA][LENGTH][PAYLOAD BYTES...] as here the frame lenth is not fixed as it was in short message.
- **Sender:** here the sender node will simply covert any long text message into bytes and then it will send the preamble followed by the length and payload same pulse encoading as we saw in short length message transmission above.
- **Receiver:** Now here the receiver will detect the preamble via sliding bytes and read the next byte as lenth then this read lenth bytes will get reconstruct the original message and print it by sending back the acknowledgment. .

3) pulse width modulation (PWM): The PWM is the technique in which we first encode the data packet by varying width of pulses and keep the frequency constant.it is like communication through timing pattern instead of the voltage levels. the different techniques used in PWM are as follow

a) PWM packet detection: here in this method we used packet detection using high blinking where a sender node repeated drives the LED with a particular high deauty pattern for the fixed duration and at receiver the it makes the samples at A0 for 10 ms over packet window of 600ms and counts the samples of the light i.e data threshold and here if count exceeds the packets light's mean value then it is treated as pwm packet and it accepts ack.

b) PWM short message detection: so here in this method we used used pwm modulation technique to transmit the short message.

- **Bit encoding:** here the bit encoding will be on pin no 3 and level 1 of the LED will be 255 for bright pulse bit 1 and for dim pulse the bit is 0 where each bit is maintained by a bit duration between 100 to 150ms following a short gap where LED is off.
- **Frame format:** here the frame format will be [PREAMBLE = 0xAA][LENGTH][PAYLOAD BYTES...]message.
- **Sender and Receiver:** here the sender node will simply send the preamble and then followed by the length of message and then 8 bits character after the frame is delivered the LED will get turned off and wait for acknowledgment and then here on the receiver side the it will use pwm specific threshold like voltage and distingush the pulses based on dim and bright.And then it will sense the delay in the middle of each threshold and then it starts building the bytes by sliding over 8 bit register untill and unless it detects preamble 0xAA.finally prints the message and sends the acknowledgment back using acknowledgment LED.

c) PWM Long Message: Here in PWM technique for long message we implemented it using the robust packetized protocol where the general idea is that a very long string is splitted into small chunks of 30 charecters and each chunk here is an separate universal frame as [preamble][length][chunk].now the node A will be using the stop and wait approach and send the chunk and wait for the acknowledgment and then move on to next.

- **Sender and Receiver:** Now here in this technique the PWM will simply use the PWM based bit wise encoding as discussed above and maintain a current index position in the long string.now here it will make a current chunk and start transmitting the frame and after transmission it will wait for particular time for the acknowledgment by sensing it from photodiode and if acknowledgment is received it moves to next frame and if not received it start retransmitting the old frame again.and now here receiver will simply start decoding each frame same like we discussed in short message technique and after verifying the length and payload it start printing the chunk and appends it to the one saparate sting from the end which is storing the entire message and finally sends back acknowledgment to node A on each successful reconstruction of decoded packet.

B. How These Methods Solve the Research Problem

Here the all of the techniques and methods discussed above solvved the different challenges differently lets see all of them

- **Multiple modulation methods (OOK, FSK, PWM)** here all these methods helped us to compare experimentally outcomes like which works the best on simple hardware and how different style of blinking affect the transmission over different distances.
- **Framing, preamble, and length field** here in these techniques the preamble tells us that receiver is about to start the new message in same way the length bit tells us how many charecters are to be used in message and frame structure makes the blinking of LED meaning ful

for transmission and all this are combined and used so that LED blinking works can be converted into proper communication packets.

- **Checksums,thresholding,packetization and acknowledgment** now here checksums and thresholding helped us to give the basic error detection and helped us to make the robust bit decision in the presence of noisy analog light reading on the other hand acknowledgment and packetization helped us to add the reliability over the unstable physical channels for both long and short text messages to be transferred correctly.

IV. Evaluation and Results

The evaluation of the system was done using the two arduino uno board placed inside the cardboard along with the entire syetem on the breadboard just to make sure that there is no disturbance of the ambient lighting in the readings of the experiment.here all the three modulations methods like OOK,PWM and FSK were tested on the same physical setup and same environment just to allow the fair comparison. Each modulation was performed for the multiple LED-photodiode distances like 0.1cm,0.2cm,0.3cm,0.4cm so that we can withdraw the conclusion via arduino serial monitor by capturing the decoding accuracy,ADC values, acknowledgment timings along with success and failure outcomes. approximately 20 to 30 trails were performed across each system for all short and long messaging techniques. and by focusing on the important parameters like bit error rate.

A. Numerical results

- **Bit Error Rate (BER)** Here all the bit error rate is nothing but the measurement of number of incorrect bits over the total number of transmitted bits in our system OOK was the modulation with extreamely reliable due to ADC difference. the pwm was somewhat stable but FSK was somewhat worst as we were transmitting the bits very slowly using LED as shown in the tables.

B. Results

The experimental evaluation compared how OOK, PWM, and FSK behave under identical hardware conditions and increasing LED–photodiode separation. For each modulation scheme, multiple message transmissions were performed at distances ranging from 0.1 cm to 0.4 cm. The receiver outputs, checksum results, and bit-level decoding accuracy were recorded to understand how quickly performance degraded as the received optical power dropped. The following tables summarize the qualitative outcome of each experiment and estimates derived from the observed decoding failures.

Among the three modulation methods, OOK consistently showed the strongest behavior across short distances. Its simple ON/OFF structure provided a clear separation between illuminated and non-illuminated ADC readings, which helped maintain very low error rates at 0.1 cm and 0.2 cm. However, even OOK began to show noticeable instability by 0.3 cm due to declining signal strength and threshold ambiguity.

TABLE I
Experimental Results for OOK Modulation

Distance (cm)	Observed Outcome	Approx. BER
0.1	All packets successfully decoded; clean threshold separation and stable link.	≈ 0
0.2	Almost all packets received; occasional packet loss but checksum usually passes.	$\approx 2 \times 10^{-2}$
0.3	Mixed performance; some frames pass checksum, others fail due to weak signal.	$\approx 1 \times 10^{-1}$
0.4	No reliable detection; LED intensity too low to cross threshold.	Not defined

PWM demonstrated a different pattern. Since PWM relies on measuring the duration of pulses rather than their amplitude, it remained workable when the LED brightness weakened slightly.The next table summarizes these behaviors

TABLE II
Experimental Results for PWM Modulation

Distance (cm)	Observed Outcome	Approx. BER
0.1	Stable decoding with clear distinction between dim and bright pulses; most frames correct.	$\approx 1 \times 10^{-1}$
0.2	Noticeable timing drift; some frames fail due to narrow margins in pulse width detection.	$\approx 3 \times 10^{-1}$
0.3	Highly unstable; only a few frames decode correctly, strong sensitivity to noise.	$\approx 5 \times 10^{-1}$
0.4	No usable decoding; dim pulses indistinguishable from background noise.	Not defined

TABLE III
Experimental Results for FSK Modulation

Distance (cm)	Observed Outcome	Approx. BER
0.1	Some frames decoded but transition counting is inconsistent; depends heavily on signal strength.	$\approx 5 \times 10^{-1}$
0.2	Very weak performance; few correct frames, transitions often not detected.	$\approx 8 \times 10^{-1}$
0.3	Almost no meaningful decoding; transition edges too weak for FSK classification.	$> 9 \times 10^{-1}$
0.4	No detection at all; FSK requires strong high-contrast pulses unavailable at this range.	Not defined

V. Conclusions and Future Work

The work presented in this project demonstrates that a fully functional Li-Fi link can be built from LEDs, photodiodes,

and Arduino boards while still supporting framed messages, checksum validation, and optical acknowledgments in both directions. By implementing and comparing OOK, PWM, and FSK on the same hardware, we were able to show experimentally that OOK offers the lowest bit error rate and most reliable operation at sub-centimeter distances, whereas PWM is more sensitive to timing drift and FSK quickly breaks down as the received optical power decreases. At the same time, the evaluation made clear that the current prototype is strongly limited by the low optical output of standard 5 mm LEDs, the lack of analog amplification or hardware comparators on the receiver side, and the use of software delays for all timing, which together restrict the usable range to roughly 0.2–0.3 cm and keep throughput in the tens of bits per second. In future work, we would like to replace the simple pull-up receiver with a transimpedance amplifier and comparator to extend the communication distance and reduce noise sensitivity, add optical elements such as lenses or collimators to better focus the LED beam, and migrate the modulation logic to hardware timers and interrupts to obtain more precise symbol timing and higher data rates. Another addition could be to introduce strong link layer mechanisms such as CRCs as well as automatic threshold adaption. This will help system to tolerate more ambient light variation and still maintain a low Bit Error Rate (BER).

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